Tool Harness Adaptation under Environment Change

PE 6203 Group Project 4 | Historical course audit, integrated research evidence | amended 2 November 2026

Research questions

Does updating affected tool knowledge recover task performance after persistent tool-environment change? Does it preserve capabilities on tasks whose tools remain unchanged? We evaluate quality, efficiency, cross-version stability, update cost and execution status separately; no weighted score is introduced.

Abstract

We implement a deterministic Selective Patch that compares tool metadata, expands the affected set through explicit dependencies, updates affected records and preserves unaffected knowledge. The same fixed DeepSeek model executes a documented three-task MCPEvol-Bench subset in Early, Middle and Late environments under No Update, Full Rewrite and Selective Patch. One execution per cell gives 27 trajectories, each with five valid judge ratings.

Independent recalculation from the frozen raw archive confirms that Selective Patch has the highest observed mean Planning Effectiveness (9.1111), but lower mean Task Fulfillment (9.8889), population ECS (9.7318) and normal completion (6/9) than both baselines. It uses zero model tokens for updating, tied with No Update, and 1.145 seconds of local update time. Its changed-task TF gain relative to No Update is 0 in Middle and +0.1 in Late; unchanged-task TF is retained.

The results establish a runnable, auditable course experiment. They do not establish general superiority, statistical significance or broad generalization. Early arms share identical knowledge, yet exhibit different truncation and completion outcomes. This variation materially affects ECS.

Later evidence ties updated parameter and route knowledge to avoided obsolete calls, while high scores or faster completion can coexist with faulty products. QI-V2 scored 25/27 planned cells without manual product certification. The 36-candidate audit found no joint completion-and-product winner. Cohorts remain separate.

Submission status

The main experiment is verified. Failure-case acceptance remains unresolved. The brief does not require naturally occurring failures. Four approved controlled diagnostics establish a missing-parameter rejection mechanism, while course acceptance of that injected case needs TA clarification. The over-update claim still lacks a demonstrated change to unrelated knowledge that causes damage. Negative diagnostics remain part of the evidence.

This offline revision adds no model calls or scoring. Original course scores, diagnostic records and unknown charges remain intact; Appendix C–E retain the historical audit. The research Goal has not passed its two-batch gate.

Evidence: results/evidence/course-v1.3-20261031/raw-trial.tar.gz; submission/evidence/recomputed/metrics.json; failure-cases.json; original course brief §§3-4.

1. Method and architecture

A harness stores tool names, input schemas, descriptions, usage rules, output schemas and routing. Persistent environment changes can make these records stale even if the base model remains fixed. Our intervention is the tool-knowledge snapshot presented to the agent, not model fine-tuning.

Adaptation pipeline

```
Early knowledge K0 + target-stage metadata
-> deterministic add / remove / field diff
-> affected records + explicit one-hop dependencies
-> copy authoritative target fields for affected records
-> preserve unaffected records and valid usage rules
-> schema / routing / preservation validation
-> frozen knowledge Kt -> common agent -> real MCP tools
-> raw trajectory + five judge ratings + independent analysis
```

Tool identity is server_id::name. The comparison covers name, parameters, description, output_schema and routing; a rename without stable identity evidence is treated as removal plus addition. Structural changes propagate through explicit tool references in usage rules. Target metadata is authoritative for structural fields. Usage rules are preserved rather than erased by an empty metadata field; unresolved semantic dependencies are recorded. The evaluated Selective Patch performs no LLM update calls.

Method	Knowledge policy
No Update	Retain the Early snapshot K0 after environmental change.
Full Rewrite	Rebuild all target records and use the fixed LLM to regenerate every target description; protect schemas, routes and supplied rules.
Selective Patch	Update affected records by deterministic diff and dependency expansion; preserve unaffected records.

All three methods use the same execution agent to choose tools, arguments and next steps. Candidate MCP servers come from the official task. No task-specific call sequence is scripted. Early uses common K0 in every arm. This implementation compares both update scope and description-generation mechanism; it is not a pure scope-only ablation.

Evidence: Frozen source src/pe6203/selective.py, baselines.py, runner.py and integration/baselines.py; per-cell adaptation/adaptation.json and knowledge.json.

2. Experimental setup and controls

Task	Role	Represented changes
133	Changed: mathematical syntax and security guidance	Middle: optional parameters; Late: required allow_variables and relevant optional tool addition.
283	Changed: analytics project decomposition	Tool additions plus parameter and description changes in planning/project tools.
235	Unchanged: supplied sales visualization	28 chart-tool records canonically unchanged; used for retention.

The fixed subset is identical across methods. TOOL coverage is limited to relevant optional additions, not mandatory renaming/removal. PARAM and DESC changes affect required tools. Metadata coverage does not mean every trajectory actually called every changed tool. Task 283 shares a service family with development task 139, so this is not an unseen-family evaluation.

The brief permits a documented fixed subset. Earlier team material considered task 223, research-paper retrieval and webpage publication; the course matrix froze executable original tasks 133/283/235 instead. It does not test the paper-webpage scenario or a verified book task. Task-223 fetch/publication services were unverified for this matrix; a later Netlify variant changes the backend. No written team sign-off for the switch was found.

Control	Frozen value
Model / provider	deepseek/deepseek-v4.1-flash-1017 via OpenRouter; same agent, judge and updater model
Decoding	temperature 0; provider seed not sent
Agent budget	4096 output tokens; 12 rounds; 900-second run budget
Timeouts / retry	request 120 s; tool 30 s; at most one HTTP retry
Updater	16384 output tokens; 120 s timeout; one HTTP retry
Judge	4096 output tokens; reasoning disabled; JSON; five first-valid slots; at most two logical attempts per slot
Measurement	Original rubric with common concise output contract and deterministic call counts appended
Execution	3 tasks x 3 stages x 3 strategies x 1 execution = 27; 135 valid rating slots

One execution is permitted by brief §3.2 under compute constraints and is explicitly disclosed. Five judge ratings are measurements of the same trajectory, not five independent agent executions. All 135 slots in this main cohort succeeded on their first logical attempt. No benchmark labels or question text were modified.

Archived checks confirm common question hashes, settings, budgets and judge rubric within matched groups. Historical cohorts stay separate.

A later registry had 29 names (27 with actor evidence, two with only generation/admission), not 29 course matrices; it later reached 36 audited and 43 total names. None of the historical 29 has task-223 actor evidence. Versions and input classes—original, fixed-fixture original, synthetic—must remain distinct. New effects need a same-condition Patch control.

3. Main results and execution status

Method	Mean TF	Mean PE	ECS pop.	ECS sample	Completed
No Update	9.9556	7.8667	9.9241	9.9171	7/9
Full Rewrite	9.9778	8.3556	9.9464	9.9393	7/9
Selective Patch	9.8889	9.1111	9.7318	9.6964	6/9

Each mean equally weights nine task-stage cells; each cell score averages five stored ratings. Scored trajectories with truncation or round exhaustion remain included. Selective Patch has observed PE gains of +1.2444 against No Update and +0.7556 against Full Rewrite. These are descriptive differences on one small cohort, not tested population effects.

Method / stage	TF	PE
No Update / Early	9.9333	8.6667
No Update / Middle	10.0000	6.6667
No Update / Late	9.9333	8.2667
Full Rewrite / Early	10.0000	9.0000
Full Rewrite / Middle	9.9333	7.8000
Full Rewrite / Late	10.0000	8.2667
Selective Patch / Early	9.6667	9.6667
Selective Patch / Middle	10.0000	9.1333
Selective Patch / Late	10.0000	8.5333

Method	Completed	Max rounds	Truncated
No Update	7	1	1
Full Rewrite	7	2	0
Selective Patch	6	2	1

Across all 27 trajectories: 20 completed, five exhausted their round budget and two were truncated. Normal completion means the agent loop ended normally with a non-empty answer, not independent proof of correctness. TF can be 10 even when the agent has not ended normally.

Evidence: evidence/recomputed/cells.csv; all run/result.json, raw_trace.json and immutable EVAL outputs in main archive.

3A. Separate QI-V2 original-task results

QI-V2 hybrid_v2 is a different Selective Patch knowledge version, pinned to bdf0ccc5c3466517725c84e7f715234733cde4a8. It planned the same 133/235/283 × Early/Middle/Late × three-method grid. FR 133-Middle and QI-V2 SP 283-Late failed during adaptation before actor execution: 25/27 cells were scored (125 valid judge slots), 19/27 ended normally. These results do not replace or pool with the complete course V1 matrix.

Method	Normal/planned	Scored/planned	Matched TF	Matched PE	Complete product
No Update	7/9	9/9	9.686	8.371	Not reviewed
Full Rewrite	6/9	8/9	9.886	8.714	Not reviewed
QI-V2 SP	6/9	8/9	9.771	7.314	Not reviewed

The TF/PE columns compare only seven triplets scored under all three methods. Available-case means use different denominators (NU 9, FR 8, QI-V2 SP 8) and are not presented as a matched effect. Missing cells remain missing, not zero; only task 235 has a complete three-stage ECS triplet for all methods. All 27 case files mark final artifact quality as not manually reviewed.

Known request cost: NU USD 0.113712221; FR USD 0.140799942; QI-V2 SP USD 0.132039527; separate preflight USD 0.000008241. FR and QI-V2 SP each retain two unknown-cost records, so these are lower bounds. Integrity replay verified all 1067 archived source files and recomputed the 125 rating slots without model calls.

Evidence: results/evidence/v2-eval-20261101/README.md, verification.json, v2-eval/summary.json and raw-results.tar.gz; archive SHA256 c36e88370426d1aedcf5368b1bf2881bcbce8e259b532c14eac3804d8b920f22.

4. Exact PE check and ECS conventions

Task	Stage	Five raw PE ratings	Cell PE
133	Early	10, 10, 10, 10, 10	10
133	Middle	9, 8, 8, 7, 8	8
133	Late	7, 7, 7, 7, 7	7
235	Early	10, 10, 10, 10, 10	10
235	Middle	10, 10, 10, 10, 10	10
235	Late	10, 10, 10, 10, 10	10
283	Early	9, 9, 9, 9, 9	9
283	Middle	9, 9, 9, 10, 10	9.4
283	Late	7, 9, 9, 9, 9	8.6

(10 + 8 + 7 + 10 + 10 + 10 + 9 + 9.4 + 8.6) / 9 = 82 / 9 = 9.111111... The reported 9.1111 is correct. The independent script uses exact rational arithmetic for score means; it never edits stored ratings.

Cross-version ECS

For each task, first compute the mean of its three TF values, then subtract their standard deviation. ECS is the average of these task-level quantities. It is not computed by subtracting the spread of three overall stage means.

```
mu_t = (TF_early + TF_middle + TF_late) / 3
sigma_population = sqrt(sum((TF_i - mu_t)^2) / 3)
sigma_sample = sqrt(sum((TF_i - mu_t)^2) / 2)
ECS = mean_over_tasks(mu_t - sigma_t)
```

The released implementation uses population standard deviation; the paper appendix specifies the sample convention. Both are reported, with population as the primary implementation-compatible result. For Selective Patch, task 133 TF=[9, 10, 10] gives task ECS 9.1953 (population) or 9.0893 (sample); tasks235 and 283 each give10. Therefore overall ECS is 9.7318 or9.6964.

The low Early TF on task 133 occurred with common K0 and output truncation. It reduces the Patch ECS, but cannot be attributed to a patch that was not yet applied. Conversely, higher subsequent TF is not automatically a pure update effect.

Evidence: evidence/recomputed/metrics.json: selective_PE and ecs_by_task; MCPEvol-Bench §6.1, Appendix D.1; upstream evaluation code in frozen source provenance.

5. Recovery and retention

Changed tasks: 133 and 283

Method	Stage	Mean TF	Delta vs same-stage No Update	Delta vs own Early
No Update	Middle	10.00	+0.00	+0.10
No Update	Late	9.90	+0.00	+0.00
Full Rewrite	Middle	9.90	-0.10	-0.10
Full Rewrite	Late	10.00	+0.10	+0.00
Selective Patch	Middle	10.00	+0.00	+0.50
Selective Patch	Late	10.00	+0.10	+0.50

Selective Patch matches No Update in Middle and is +0.1 TF higher in Late on the two changed tasks. Its own Early-to-Late increase is partly sensitive to the Early truncation. The more direct local mechanism is task 133 Late: four initial No Update calls omit a newly mandatory parameter and are rejected; Patch provides that parameter and has no tool errors in its corresponding run. No Update later repairs its calls using error feedback, reducing the final TF gap.

Unchanged task: 235

Method	Stage	TF	PE	Delta PE vs own Early
No Update	Middle	10.0	4.8	-5.2
No Update	Late	10.0	10.0	+0.0
Full Rewrite	Middle	10.0	10.0	+0.0
Full Rewrite	Late	10.0	10.0	+0.0
Selective Patch	Middle	10.0	10.0	+0.0
Selective Patch	Late	10.0	10.0	+0.0

Every task 235 TF is 10, so all TF retention deltas are zero. Patch retains its three chart-tool calls and PE 10. No Update Middle has five map calls and PE 4.8 despite unchanged knowledge, showing that redundant actions need not come from rewriting. Tool output consists of chart specifications: browser rendering and a hosted dashboard were not validated.

Evidence: Recomputed recovery and retention arrays; tasks133/283 labelled changed and 235 unaffected before scoring in the frozen suite.

5A. Follow-up mechanisms and results

Two separate original-task migrations show where tool knowledge can help. In task 90 Late, NU called an obsolete entry twice and failed to retrieve the current daily problem, while updated methods used get_problem(daily=true); all four methods nevertheless completed all three stages. In task 430 Late, NU attempted deleted search_symbols seven times; original SP used browse_module(query) successfully. FR and budget_feedback avoided the old call but did not exercise that new query route. Across the task-430 cohort, SP and budget_feedback completed 3/3, NU and FR 1/3, while all four mean TF scores were 9.9333. This establishes local route recovery, not complete Swift-deliverable correctness.

Distinct cohort	SP normal	Candidate normal	All-property product	Reading
sufficiency_feedback confirmation	17/27	14/27	Not established	PE improved
observed_contract, changed, two batches	8/12	5/12	0/12 each	Validation failed
budget_feedback, mixed local six	2/6	6/6	Code/numeric defects	Lead only

The 108-trajectory sufficiency confirmation raised changed-stage PE from 8.5444 to 9.4556, yet changed-stage completion tied at 11/18; the all-stage gap arose in Early when knowledge was identical. Observed_contract finished 3/6 then 2/6 changed tasks versus SP 4/6 in each validation batch; neither arm produced a complete product in those 12 changed-task runs. Budget_feedback's local completion gain includes reused/mixed conditions: one 283-Middle run saved a 36-record, 356-hour plan, but other answers contained a SyntaxError or wrong annual total. These observations support narrower mechanisms, not a transferable joint win.

The offline audit covers 36 registered candidates across 45 cohorts, 517 executed cells, 2555 valid judge slots and 445 existing product reviews; seven recent rounds contain 69 actor trajectories, six complete rounds and one interrupted round. No candidate satisfies strict normal-completion and complete-product gains with TF and retention safeguards. The counts describe research maturity, not comparable effect sizes across differing tasks and fixtures.

Evidence: results/evidence/daily-migration-pilot-20261031; tool-migration-pilot-20261031; selective-search-20261031/confirmation-r1-review.json; observed-contract-validation-20261031/validation-verification.json; matched-existing-20261031/trial-verification.json and quality-review.json; results/evaluation/current-branch-5b772ff-20261102/REPORT.md.

6. Update and execution costs

Method	Update tokens	Update seconds	Update API USD
No Update	0	0.000000	0.000000000
Full Rewrite	177974	203.507975	0.049176082
Selective Patch	0	1.145192	0.000000000

Selective Patch is cheaper to update than Full Rewrite, but is tied with No Update at zero model tokens and zero API dollars. No Update also has zero measured update time, whereas Patch performs local diff and validation work. Zero API dollars does not imply zero computation or engineering cost.

Main-cohort cost component	Known USD
Agent requests	0.278473878
Judge requests	0.079262222
Adaptation	0.049176082
Preflight	0.000007110
Known total (lower bound)	0.406919292

Unknown billing: task 283 Late Selective Patch had an APIConnectionError at agent round 5 followed by the preconfigured retry. The failed request has no usage or fee data. Recorded aggregate tokens and complete USD therefore remain null for that agent execution; successful-response usage supplies a known lower bound only. The total above is not an exact invoice.

Update token/API amounts were checked against raw updater response usage; update elapsed time comes from stored adaptation timer records. Agent and judge costs were recomputed from their response usage, including failed-request uncertainty. Local timings are not independent wall-clock measurements. Update costs are not multiplied by the five judge ratings.

Historical development, diagnostics and confirmation runs are separate costs. Six retention diagnostics cost a recorded $0.0527461802; the main-cohort known total excludes them. Failed confirmation rounds and stopped connectivity diagnostics retain their own logs and incomplete bills in the historical archive set.

Evidence: evidence/recomputed/metrics.json costs and unknown_billing; adaptation/updater-audit.json; raw trace request_retry event; history archive manifest.

7. Failure evidence: under-update and confounds

We classify a stated claim as **confirmed** (direct mechanism evidence), **suggestive** (plausible but confounded), **inconclusive** (insufficient evidence), or **contradicted** (inconsistent with stored knowledge/calls). Confirmation of a local failure does not establish a general strategy effect.

U1 - Confirmed stale No Update call failure

Task 133 Late: Early validate_syntax requires expression only; Late also requires allow_variables. No Update keeps the Early schema. Four initial validation calls omit allow_variables and are rejected; later calls add the parameter after reading errors. The agent completes with TF 9.8/PE 7.2 after 38 calls. The stale-schema-to-rejection chain is direct, but it is a No Update observation, not a Selective Patch under-update example.

U2 - Contradicted omission attribution for Patch

The main Selective Patch artifact contains the required field and its three validate_syntax calls explicitly pass allow_variables. It completes with TF 10/PE 7, four total calls and zero tool errors. Calling this a missed parameter update would contradict the actual records.

B1 and P1 - Different failure mechanisms

Task 133 Early Patch truncates in round 2 with TF 9/PE 10, although every Early arm has identical K0. An update-caused explanation is contradicted. Task 283 Late Patch reports review_workspace_health failing because listMemories is missing; it also has two empty-argument create_task errors, an API retry and round exhaustion. The tool error is observed directly, but its contribution to the terminal failure is not isolated. Neither should automatically be called under-update or over-update.

Audit scope

We screened all 317 accessible raw_trace.json files, with comparable old/updated/target metadata for 298. The 72 structural candidates comprise 31 No Update stale snapshots, 40 Full Rewrite cases with edits to metadata-unchanged tools, and one linked_notes case. Six stale-parameter mechanisms are confirmed, two cases are suggestive and 64 remain inconclusive. Nine selected case studies receive deeper semantic review. The screen does not prove absence of all semantic usage-rule omissions; 19 traces lack complete comparable metadata.

Controlled evidence: removing the required allow_variables fact from a correct snapshot produces eight matching rejections, versus zero with the complete snapshot. This confirms the local injected mechanism. It does not show that the original updater naturally omitted the field. The brief does not impose a natural-occurrence condition. Whether this controlled trajectory satisfies the failure-analysis deliverable remains a TA question, not an automatic pass or fail. See Appendix E.

Evidence: evidence/trajectory-inventory.csv; trajectory-screening.json; failure-cases.json; case-details/U1.json, U2.json, B1.json, P1.json. Each case includes exact old/updated/target fields and calls.

8. Failure evidence: over-update and negative results

O1 - Suggestive historical Full Rewrite regression

In the older task 235 Late cohort, all 28 chart records are unchanged across environments. Full Rewrite nevertheless regenerates every description. Its trajectory makes five map calls and omits map-unit metadata, with TF 8.8/PE 4.2 versus 10/10 for the other arms. However, the rewritten description explicitly mentions dataValueUnit. The omission therefore cannot be explained as guidance simply disappearing. The regression is not itself proof that rewriting caused damage, and this is Full Rewrite rather than Selective Patch.

O3 - Inconclusive causal attribution after negative diagnostics

Six existing controlled description-rollback executions, three per arm, all complete with TF/PE 10 and three tool calls. They do not reproduce the historical damage. We retain this negative result and do not repeat it until a desired effect appears. It does not establish that harmful rewrites are impossible.

O2 - Suggestive scope expansion in linked_notes

A historical research variant adds this rule to metadata-unchanged math-mcp::evaluate:

validate_syntax now accepts optional detailed and allow_variables flags; evaluate's validate_only remains a separate syntax-only path, so choose based on whether evaluation is also wanted.

The task 133 Middle trajectory performs two syntax checks and a security-prompt call, then three evaluate calls. It completes with TF 10/PE 5; original Patch in that cohort uses three calls and scores 10/10. Four usage-rule records differ between these arms, so the note is not an isolated treatment. Also, evaluate is semantically related to the changed validator: metadata-unchanged does not necessarily mean unrelated. This is a suggestive research-variant case, not a confirmed failure of the main method.

Judge limitation: correct scores are not corrected answers

A fresh factual review of the six diagnostic answers finds three claiming the monthly sales sum is 24.4M: rewrite-r0, rewrite-r2 and rollback_descriptions-r0. The supplied 12 numbers sum to 25.4M. All three retain their original TF/PE 10. Errors in both arms weaken any claim of rewrite-specific factual harm and demonstrate that high judge ratings can miss answer defects.

Unresolved over-update evidence: a later one-rule retained/reverted comparison is suggestive, but the rule concerns a related validator and cannot be assumed unrelated. Thus neither that pair nor the negative chart rollback establishes the full unrelated-change-to-damage chain required by the definition. See Appendix E. Lack of natural occurrence is not the acceptance criterion.

Evidence: case-details/O1.json and O2.json contain exact descriptions/rules; diagnostic-review.json identifies all six trajectories; raw answers and five ratings remain unchanged.

9. Interpretation and limits

Supported observations

Selective Patch updates tool knowledge without model calls and prevents the specific initial missing-parameter errors seen in No Update task 133 Late. In this cohort it retains task 235 TF and has the highest mean PE. Full Rewrite leads the mean TF and ECS comparisons; both baselines have more normally completed runs. These outcomes can coexist and do not require a composite winner.

The successful local mechanism is specific: an authoritative parameter diff made task-133 Late validation calls admissible before error-driven repair, while preserved task-235 chart records retained the observed specification behavior; later original-task migrations show that updated routing can recover a moved lookup. These facts explain avoided stale calls and recovered information. They do not by themselves explain the course mean PE difference or prove complete final artifacts.

What the data do not establish

One execution per cell on three tasks is not enough to estimate robust strategy-level uncertainty, frequency of rare failures or generalization to the full benchmark. Temperature0 does not make provider execution deterministic. Five judge ratings characterize one saved trajectory, not run-to-run agent variance. No inferential significance claim is made for the27-run main cohort.

TF is near the scale ceiling, and incomplete trajectories can receive high TF. Early differences under identical K0 affect cross-version scores. The same model family acts as agent and judge. Rubric interpretation of distinct security checks and redundant calls is imperfect; manual factual findings are reported separately, never silently substituted for ratings.

The subset lacks required tool removal/renaming, uses optional tool additions, and is not an unseen-family holdout. The runner uses native stdio and common local adaptation plumbing, rather than an exact recreation of every paper infrastructure choice. Full Rewrite and Patch differ in both update scope and generation mechanism. Public MCP package defects are kept common across arms.

The follow-up cohorts locate the remaining gap: fewer calls, higher PE, better individual explanations or early completion did not consistently produce runnable code, correct totals and a delivered answer together. QI-V2 has missing adaptation cells and no manual all-property review. None of these cohorts can be used to declare the original Patch superseded, and synthetic diagnostics remain mechanistic checks rather than original-benchmark validation.

Research decision

The course comparison is complete for its stated subset; the stronger research Goal remains unmet. Observed_contract failed both validation batches, and no later version earned promotion. A future frozen mechanism needs a contemporaneous Patch control and strict gains in normal completion and every required product attribute, with TF and retention preserved, in development and separately in two independent batches. Failures and unknown charges remain recorded; task 223 and synthetic fixtures cannot silently replace the frozen subset.

10. Reproducibility and delivery

The repository delivery contains this updated report, shared presentation and current notes, independent calculations, frozen source/configuration and raw archives. The original report and two ZIP bundles remain frozen historical snapshots. This revision supersedes their earlier strict-natural acceptance wording without changing any scores. HANDOVER.md tracks open items and the Git delivery receipt. No external deployment or TA message is sent.

Offline recalculation - no credentials or dependencies

```
python3 submission/handover-20261031/verify.py
```

From the repository root, run: python3 submission/handover-20261031/verify.py. It blocks networking, recomputes the main 27 cells and 135 ratings, checks all target-record rewrites and actual changed-tool calls, reconstructs10, 959 frozen files from committed archives in a temporary directory, replays the historical case audit and independently recalculates the four diagnostic cells. It requires only Python3.12+ standard library. Inputs remain read-only; the experimental pipeline is not executed.

Future execution under a new frozen protocol

```
mkdir -p frozen-source
tar -xzf submission/evidence/source-main-12d693c.tar.gz -C frozen-source
cd frozen-source
uv sync --frozen
# Install pinned Node and released MCP environments per RUNBOOK.html.
# Configure credentials locally; do not put them in the package.
./experiment prepare --suite experiments/benchmark-v1.3.json \
--repeats 1 --output results/raw/approved-new-run
./experiment run --output results/raw/approved-new-run
```

These commands document the historical environment, not a reason to rerun completed cells. The user has authorized bounded local research, but any new experiment must freeze its distinct mechanism, task conditions, product acceptance, schedule and ledger before model calls. Archived arithmetic is exactly reproducible; a fresh behavioral run is billable and may differ. No push, deployment or external message is part of this report revision.

Provenance

Main source commit: 12d693c. Benchmark commit: 888c779a0fe735f612ca62bdd1fdd7c9e6bf8701. Dataset revision: 162e3ffbea1717ac8a20fe9b26465e29117229e4. Runtime recorded as macOS arm64, Python3.12.12, Node 24.13.0; Python dependencies are locked in uv.lock. Full released environment archives require a separate large download; the raw arithmetic replay does not.

```
Main archive SHA-256:
e95e15694abe9a12e37c813f81e6e0fb0b5ccd00c2003f2a25a36a092c8001c9
```

References

[1] PE 6203 Group Project 4: Tool Harness Adaptation under Environment Change, course brief, §§3-4. [2] Liu et al., MCPEvol-Bench: Benchmarking LLM Agent Performance Across Dynamic Evolutions of MCP Servers, arXiv: 2607.14642v1, 2026, §6.1 and AppendixD.1. https://arxiv.org/html/2607.14642v1 . [3] Frozen benchmark release and course implementation identified above. [4] Local immutable experiment and diagnostic records indexed in evidence/.

Appendix A. All 27 scored cells

Task	Stage	Method	TF	PE	Execution status
133	early	No Update	9.8	8	output_truncated
133	early	Full Rewrite	10	8	completed
133	early	Selective Patch	9	10	output_truncated
133	middle	No Update	10	7.8	completed
133	middle	Full Rewrite	9.8	5.6	completed
133	middle	Selective Patch	10	8	completed
133	late	No Update	9.8	7.2	completed
133	late	Full Rewrite	10	5.8	completed
133	late	Selective Patch	10	7	completed
235	early	No Update	10	10	completed
235	early	Full Rewrite	10	10	completed
235	early	Selective Patch	10	10	completed
235	middle	No Update	10	4.8	completed
235	middle	Full Rewrite	10	10	completed
235	middle	Selective Patch	10	10	completed
235	late	No Update	10	10	completed
235	late	Full Rewrite	10	10	completed
235	late	Selective Patch	10	10	completed
283	early	No Update	10	8	completed
283	early	Full Rewrite	10	9	max_rounds
283	early	Selective Patch	10	9	completed
283	middle	No Update	10	7.4	completed
283	middle	Full Rewrite	10	7.8	completed
283	middle	Selective Patch	10	9.4	max_rounds
283	late	No Update	10	7.6	max_rounds
283	late	Full Rewrite	10	9	max_rounds
283	late	Selective Patch	10	8.6	max_rounds

Every row averages five valid ratings. Status is retained independently of the scores. Raw rating arrays, exact fractions, tool-call counts and per-cell usage are available in evidence/recomputed/metrics.json and cells.csv.

Appendix B. Completion checklist

Required deliverable	State	Evidence / outstanding work
Selective method and two baselines	Ready	Frozen source, configs and 27 real trajectories.
TF/PE/ECS and execution status	Verified	Independent raw-archive recalculation; both ECS conventions.
Recovery and retention	Verified, limited scope	Two changed tasks and one unchanged task; no broad claim.
Update cost	Verified with uncertainty	Tokens/time/API separated; one main failed-request charge unknown.
Under-update failure trajectory	TA acceptance pending	Controlled omission confirms local rejection; original updater did not omit the field.
Over-update failure trajectory	Evidence incomplete	Related-rule contrast is suggestive; unrelated-change causal damage unproven.
Report and reproducibility	Prepared for review	This report, scripts, logs, source and manifests.
10-minute presentation + Q&A	Prepared for review	Editable deck, timed speaker notes and Q&A; group rehearsal outstanding.
Final submission	Repository delivery prepared	Git status in HANDOVER / delivery receipt; team course registration and TA clarification remain.

The brief requires example trajectories with under-update and over-update analysis, but does not require natural occurrence. An injected omission is labelled as such; it is not automatically invalid or sufficient. Over-update still needs evidence that changed knowledge was unrelated and caused harm. Course acceptance and Full Rewrite interpretation need TA clarification. This report does not declare those unresolved items complete.

The next permissible work is offline review and group preparation. Additional model calls, diagnostics, task/repeat expansion and Netlify deployment remain disallowed without explicit approval.

Appendix C. Full Rewrite scope audit

Brief §§2 and 4.1 call for rewriting the complete tool-related knowledge. The implemented baseline rebuilds every record from target metadata and asks the fixed LLM to regenerate all target descriptions. This is full-catalog regeneration under a protected-structure interpretation, not free-form generation of every field. The archived implementation is unchanged.

Knowledge field	Observed handling	Evidence / limit
Tool membership, identity, name	Rebuild the complete target list; additions included and obsolete identities removed	All six updated snapshots cover exactly the target identities.
Input schema / parameters	Copy target JSON Schema unchanged	LLM is not permitted to change schema or constraints.
Description	LLM regenerates every target description, including unchanged tools	209/209 returned and textually different from target prose.
Usage rules	Copy supplied target rules verbatim; absent becomes []	Validator rejects invented or changed rules; does not regenerate independent rules.
Output schema	Copy target field, including null when unavailable	Same representation reaches agent across all methods.
Routing / server IDs	Copy target routing; shared runner applies deterministic namespacing	No LLM routing synthesis.
Early condition	Reuse identical K0; no updater call	Three Early controls are not counted as rewrites.

Task	Stage	Target records	Descriptions regenerated
133	middle	43	43
133	late	45	45
235	middle	28	28
235	late	28	28
283	middle	31	31
283	late	34	34

The updater receives only the target catalog and constraint checks, not old knowledge, task text, answers, scores or evolution labels. All three methods share the same model, decoding, task, execution budget and judge protocol. Their update algorithms intentionally differ: LLM description regeneration versus deterministic affected-record copying. Therefore results compare methods, not a controlled scope-only ablation. Schema preservation avoids inventing incompatible APIs but narrows the meaning of Full Rewrite. Ask the TA whether this operationalization meets the expected complete-knowledge baseline.

Evidence: handover-20261031/audit.json full_rewrite; archived adaptation/updater-audit.json requests and responses; frozen integration/baselines.py and src/pe6203/baselines.py at 12d693c.

Appendix D. Change coverage and actual contact

Each transition below compares Early K0 with the stated target environment, matching how adaptation was executed. Counts are actual model tool calls in the main 27 trajectories, ordered No Update / Full Rewrite / Patch. A changed catalog entry, task relevance and executed contact are different evidence levels. Contact does not isolate the behavioral effect of one changed description.

Type / task / target	Environment and task relevance	Actual calls NU / FR / SP
PARAM /133 /Middle	validate_syntax adds optional detailed and allow_variables; central to expression checking	6 /12 /11
PARAM /133 /Late	validate_syntax requires allow_variables; direct required-tool change	13 /5 /3. Initial NU omissions rejected; Patch supplies parameter.
DESC + PARAM /283 /Middle, Late	create_project adds metadata parameters and changes description; directly relevant project setup	1 /1 /1 in each stage. Optional fields need not all be used.
TOOL /283 /Late	review_workspace_health newly available; optional project-workspace inspection, not required by task	0 /1 /1. FR succeeds with includeMemories=false; Patch errors with true.
TOOL /133 /Late	batch_evaluate addition is relevant to multi-expression work, optional	0 /0 /0 in main cohort. Used in controlled diagnostic only.
TOOL /283 /Middle, Late	parallel-thinking / convergent-synthesis planning additions, optional	0 /0 /0 for these new tools.
DESC (+ PARAM Late) /283	sequentialthinking changes; relevant reasoning capability	0 /0 /0 in Middle and Late.
Retention /235 /all stages	All 28 chart records unchanged after removing version tags	No TOOL / PARAM / DESC change; retention control.

The observed review_workspace_health failure is Error: memoryStorage.listMemories is not a function. The differing includeMemories arguments expose a common upstream defect; they do not establish a patch-update error. No Update does not know this new tool. The tool is optional and task relevance is an analyst interpretation based on project/workspace planning, not a new required-tool label.

Thus the subset contains all three change levels and includes executed contact for each. Coverage remains narrow: optional additions only, no required deletion/rename, and no isolated description-only causal test. Other server-catalog changes unrelated to the task are not counted as stronger task coverage. The complete 29 changed records, exact field differences, arguments and observations are retained in the audit JSON and coverage CSV.

Evidence: handover-20261031/coverage.csv; audit.json coverage; main archive catalogs/{task}/{stage}.json and cells/task{task}-{stage}-{method}-r0/run/raw_trace.json.

Appendix E. Approved diagnostics and TA questions

Controlled condition	TF	PE	Calls / errors	Terminal state
Under: omitted fact	10	5.8	27 /9	Truncated
Under: complete fact	10	5.6	5 /1	Completed
Over candidate: rule retained	10	7.8	22 /2	Truncated
Over candidate: rule reverted	9.8	10	4 /0	Completed

Under-update mechanism: confirmed fault injection. Task 133 Late removes only allow_variables from validator properties and required fields. Eight calls omit it and receive the exact missing-field rejection; the complete arm has zero such rejections. Both TF values remain10 and PE does not improve on correction. Another tool implementation error and four incorrect judge rationales affect the complete arm. This is a controlled knowledge omission, not evidence that the original adaptation algorithm missed the change.

Over-update candidate: suggestive, definition not established. Task 133 Middle uses a historical linked_notes snapshot. Only one evaluate usage rule differs between retained/reverted arms; the other three historical rules are held fixed. The rule discusses the changed validator and the distinction between validation and evaluation. Consequently metadata-unchanged evaluate is not demonstrably unrelated knowledge. More calls, lower PE and truncation coexist with the rule, but one stochastic execution per arm and a TF tradeoff prevent a strong causal claim. Earlier six chart rollback runs remain negative.

All four runs were previously approved, used the main frozen model/budgets/judge and no updater calls, and cost $0.04600192 with no unknown charges. Recorded execution plus judging time is 160.707 seconds. They are separate from the27-cell main means. This final audit adds zero model calls and does not replace any score.

Draft clarification questions - not sent

1. May the explicitly injected missing-knowledge trajectory satisfy under-update analysis despite unchanged TF and no demonstrated original-updater omission? The brief contains no natural-occurrence requirement. 2. May we present over-update candidates and negative controls as an unresolved finding, or is a stronger unrelated-change failure required? 3. Does full-catalog rebuilding with all descriptions regenerated and schema/rules/routing copied from authoritative target metadata meet the Full Rewrite definition? 4. Does the documented optional-addition coverage with actual contact meet the intended Tool-level scope, with deletion/rename untested?

TA clarification remains open. Later user authorization permits bounded local research; frozen scores are unchanged, and external submission or deployment is not authorized.

Evidence: diagnostics-20261031/plan.json, exact-knowledge-diffs.json and audit.json; targeted-diagnostics archive; HANDOVER.md. This clarification supersedes earlier strict-natural wording; diagnostic facts remain unchanged.